



MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust®, Mangalore)

Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi

Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution

FakeXpose

MINI-PROJECT

(25MCSE516)

REPORT

Submitted by:

Karthik Pai

In partial fulfillment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

2025-26



MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust®, Mangalore)

Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi
Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution

DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

CERTIFICATE

This is to certify that **Karthik Pai** has successfully completed his second semester Mini-Project entitled **FakeXpose** as a partial fulfillment of the requirements for the award of **MASTER OF COMPUTER APPLICATIONS** degree, during the Academic Year **2025-26** under my supervision. This report has not been submitted to any other Organization/University for any award of degree.

Signature of the Guide

Head of the Department

Semester End Examination

Examiner 1

Examiner 2

Name :

Name :

Signature :

Signature :

DECLARATION

I, **Karthik Pai**, student of II Semester MCA, bearing
hereby declare that the project work entitled “**FakeXpose**” has been carried out by
me under the supervision of an Internal Guide **Mr. Shylesh B. C**, Assistant
Professor, Department of MCA and submitted in partial fulfillment of the
requirements for the award of the Degree of **Master of Computer Applications**,
Mangalore Institute of Technology & Engineering, an Autonomous Institution,
Affiliated to Visvesvaraya Technological University during the academic year **2025-**
26. This report has not been submitted to any other Organization/University for any
award of degree.

Name :

Signature:

Date:

ACKNOWLEDGEMENT

I would like to thank **Mr. Rajesh Chouta**, Chairman, Mangalore Institute of Technology & Engineering for providing a good infrastructure for timely completion of my mini project.

I sincerely express my gratitude to **Dr. Prashanth C M**, Principal, MITE, for supporting me in all my technical activities and giving proper guidance when it was required. I would like to thank **Dr. Madhwaraj K G**, Professor & HOD, Department of MCA, for granting permission to undertake this project. I would like to express my sincere appreciation to our project guide **Mr. Shylesh B. C**, Assistant Professor, Department of MCA for providing timely instructions and valuable guidance at every stage of the Project work.

I thank all the staff members of the Department of Master of Computer Applications, for extending their constant support to complete this project. I express my heartfelt thanks to my parents and friends who were also a constant source of support and inspiration throughout this project.

TABLE OF CONTENTS

Chapter No.	Title	PageNo.
	ABSTRACT	
1	INTRODUCTION	
1.1	General Introduction	
1.2	Objectives of the project	
1.3	Existing System	
1.4	Proposed System	
1.5	Feasibility Study	
		1-5
2	REVIEW OF LITERATURE	
2.1	Review–Summary	
		6
3	SYSTEM CONFIGURATION	
3.1	Hardware Requirements	
3.2	Software Requirements	
		7
4	MODULE DESCRIPTION	
4.1	Modules	
		8-9
5	SYSTEM DESIGN	
5.1	Data Flow Diagrams	
5.2	Use Case Diagram	
		10-12
6	SYSTEM IMPLEMENTATION	
6.1	Implementation	
6.2	Screenshots	
		13-16

7	SYSTEM TESTING	
7.1	Test Cases	17-20
8	RESULTS AND DISCUSSION	
8.1	Conclusions	21
8.2	Future Enhancements	
9	REFERENCES	
9.1	Text References	22
9.2	Web References	

LIST OF FIGURES

Sl. No.	Figure No.	Title	Page No.
1	5.1.1	Data Flow Diagram: Level 0	10
2	5.1.2	Data Flow Diagram: Level 1	11
3	5.2.1	Use Case Diagram	12
4	6.2.1	Project Home Page	14
5	6.2.2	Signup Page	14
6	6.2.3	Login Page	15
7	6.2.4	Fake News Detection[Verified News]	15
8	6.2.5	Fake News Detection [Unverified News]	16
9	6.2.6	Fake News Detection [False News]	16

ABSTRACT

The Project aims to develop an AI-powered Fake News Detection and Verification System that analyzes user-provided news or claims and determines whether they are Likely Verified, Likely False, or Unverified. Unlike traditional fake news detectors that rely only on machine learning models or Large Language Models (LLMs), this system verifies information using trusted news sources and fact-checking services before making a decision.

When a user enters a news article or claim, the system first uses the Gemini API, Google Search API and News API to extract important information such as keywords, named entities, and factual claims. Gemini is used only to understand the input and generate explanations; it does not decide whether the news is true or false. The extracted information is then used to search for related news articles through News APIs and verified fact-checking databases.

The retrieved articles are compared with the user's claim using Sentence Transformers (SBERT), which measures the similarity in meaning between the claim and the available news articles. The system also evaluates the credibility of each source by assigning higher trust scores to reliable organizations such as government websites and well-known news agencies, while giving lower scores to unknown or unreliable websites. Social media content is treated only as an early indicator of trending claims and is not considered as direct evidence.

The system combines information from semantic similarity analysis, trusted news sources, source credibility, and fact-check results using a rule-based decision engine. Based on the collected evidence, it classifies the claim as Likely Verified, Likely False, or Unverified. Finally, the Gemini API generates a simple explanation describing why the claim received its classification and what evidence was used to support the result.

CHAPTER 1

INTRODUCTION

1.1 General Introduction

Fake news has become a major problem due to the rapid growth of the internet and social media. False information spreads quickly and can mislead people, create panic, influence public opinion, and even affect political, economic, and social stability. With millions of users sharing news through websites, social media platforms, blogs, and messaging applications, it has become increasingly difficult to distinguish between genuine and misleading information. The rapid spread of misinformation can damage public trust, create confusion, and lead to poor decision-making. Therefore, there is a growing need for an intelligent system that can verify the authenticity of news before it is accepted or shared by users.

Traditional fake news detection methods mainly rely on machine learning models trained on historical datasets. These models classify news as either real or fake based on patterns learned from previously available data. Although such approaches achieve good accuracy on known datasets, they often fail to identify newly or breaking news because they cannot access current information from the internet. As a result, these systems may produce incorrect predictions when faced with recent events or claims that were not present during training.

Recent advances in Artificial Intelligence, especially Large Language Models (LLMs) such as Gemini, have significantly improved the ability of computers to understand and process natural language. These models can summarize text, identify important entities, and extract meaningful information from user input. However, relying only on an LLM to determine whether a news article is true or false is not sufficient. Since LLMs generate responses based on learned patterns and may not always verify information using live or trusted sources, they can sometimes produce inaccurate, outdated, or hallucinated answers.

To overcome these limitations, this project proposes an AI-powered Fake News Detection and Verification System that combines artificial intelligence with evidence-based verification techniques. Instead of depending only on machine learning predictions, the system first extracts important claims, keywords, and entities from the user's input using the Gemini API. It then

searches trusted news sources through News APIs and checks fact-checking organizations using Fact Check APIs. The retrieved articles

are compared with the user's claim using Sentence Transformers (SBERT) to measure semantic similarity, while a source credibility analysis evaluates the reliability of the news sources.

The system also includes a rule-based decision engine that combines semantic similarity scores, source credibility scores, and fact-checking results to classify the news as Likely Verified, Likely False, or Unverified. Finally, Gemini generates a simple explanation describing why the claim received its classification and what evidence was used during verification.

1.2 Objectives of the Project

The main objective of this project is to develop an intelligent and reliable Fake News Detection and Verification System that helps users determine the authenticity of news articles and claims. The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) to analyze the input provided by the user.

It extracts important claims using the Gemini API, retrieves related information from trusted news sources, and compares the claim with retrieved articles using Sentence Transformers (SBERT) for semantic similarity analysis.

The system further verifies the claim using professional fact-checking services and evaluates the credibility of the news sources. Based on the collected evidence, a rule-based decision engine classifies the news as Likely Verified, Likely False, or Unverified.

Finally, the system provides a clear explanation of the verification result, enabling users to understand why a particular decision was made and helping them make informed decisions before believing or sharing online information.

The objectives of this project are:

- To detect fake news using AI and NLP.
- To extract claims from user input using Gemini.
- To retrieve related news from trusted sources.
- To compare news using SBERT semantic similarity.

- To verify claims using fact-checking services.
- To classify news as Likely Verified, Likely False, or Unverified.
- To provide users with an explanation of the result

1.3 Existing System:

Most existing fake news detection systems rely on machine learning algorithms or artificial intelligence models that are trained using previously collected datasets. [1] These systems classify news articles as either real or fake by learning patterns from historical data. but they achieve good accuracy on known datasets, they often struggle to detect newly emerging or breaking news because they cannot access or verify real-time information available on the internet.

Many existing systems do not compare user claims with current news articles published by trusted media organizations or verify them using professional fact-checking services. [2] As a result, they may produce incorrect or misleading predictions when faced with recent events, viral social media posts, or claims that were not included in the training data.

In addition, some AI-based systems rely entirely on language models to make decisions, which can lead to inaccurate or hallucinated responses due to the lack of real-world evidence. Another major limitation is that these systems usually provide only a final prediction, such as "Fake" or "Real," without explaining the reason behind the decision or presenting supporting evidence. This lack of transparency makes it difficult for users to trust the results and understand how the system reached its conclusion.

1.4 Proposed System:

The proposed system introduces an AI-powered Fake News Detection and Verification System that verifies news using real-world evidence instead of relying only on artificial intelligence predictions. Unlike traditional systems that depend mainly on machine learning models trained on historical datasets, the proposed system combines AI with live information from trusted sources to provide more accurate and reliable verification.

When a user submits a news article or claim, the Gemini API first analyzes the text and extracts important claims, keywords, and named entities. These extracted claims are then used to search for

relevant information through News APIs, while Fact Check APIs are used to determine whether the claim has already been verified by professional fact-checking organizations.

The retrieved news articles are compared with the user's claim using Sentence Transformers (SBERT), which calculates the semantic similarity between the submitted claim and the available evidence. In addition, the system evaluates the credibility of the news sources by assigning trust scores to trusted media organizations, government websites, and verified fact-checking platforms.

A rule-based decision engine combines the semantic similarity score, source credibility score, and fact-checking results to determine the final classification of the news. Instead of simply labeling the news as "Fake" or "Real," the system classifies it as Likely Verified, Likely False, or Unverified, depending on the available evidence.

Finally, the Gemini API generates a clear and user-friendly explanation describing the reasons for the decision and the evidence used during verification. This evidence-based approach improves the accuracy, transparency, and reliability of fake news detection while helping users make informed decisions before believing or sharing online information.

1.5 Feasibility Study

The feasibility study determines whether the proposed system can be successfully developed and implemented. The project is technically, economically, operationally, and practically feasible.

Technical Feasibility

The system is developed using Python, Flask/FastAPI, HTML, CSS, JavaScript, Gemini API, News APIs, and Sentence Transformers. These technologies are widely available and suitable for building an AI-powered web application.

Economic Feasibility

The project requires minimal development cost because most software tools, libraries, and frameworks are open source. Free API plans can also be used during development.

Operational Feasibility

The application provides a simple and user-friendly interface. Users only need to enter a news article or claim, and the system automatically verifies it and displays the result with an explanation.

Time Feasibility

The project can be completed within the planned academic schedule using available tools, APIs, and open-source technologies. Proper planning and testing ensure successful implementation within the project timeline.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Review of Literature

Several researchers have proposed different methods to detect fake news using Machine Learning and Natural Language Processing (NLP). One research paper, "Fake News Detection using Machine Learning: Integrating FakeBERT Classification, Style Analysis, and Credibility Verification" [3] (2025) by Rafael Nicholas Tanaja, Johnny, Musyaffa Ayman Rafif, and Alexander Agung Santoso Gunawan, uses FakeBERT to classify news articles. The system also performs style analysis, credibility verification, and TF-IDF with cosine similarity to compare news with trusted sources. a Random Forest classifier is also used to classify the news as fake or real. This approach improves accuracy by combining multiple techniques.

Another research paper, "Fake News Detection using Machine Learning and NLP with Missing Data Handling" [4] (2023), proposes a fake news detection system using Natural Language Processing (NLP) and Machine Learning. The system performs text preprocessing, Part-of-Speech (POS) tagging, sentiment analysis, TF-IDF feature extraction, and missing data handling. It uses Naïve Bayes, Passive Aggressive Classifier, and Deep Neural Network (DNN) models to classify news articles as fake or real. The paper shows that combining NLP techniques with machine learning improves the performance of fake news detection.

These research papers provide good accuracy, they mainly depend on trained datasets and machine learning models. They do not fully verify claims using real-time news sources, trusted fact-checking websites, or evidence-based verification. From the literature review, it is observed that existing fake news detection systems mainly rely on machine learning models and NLP techniques for classification. Most systems use offline datasets and classify news as either fake or real without verifying current information from trusted sources.

Our system improves upon these approaches by using Gemini API for claim extraction, News APIs for retrieving real-time news, Sentence Transformers (SBERT) for semantic similarity, fact-checking services, source credibility analysis, and a rule-based decision engine and classifies them as Likely Verified, Likely False, or Unverified.

CHAPTER 3

SYSTEM CONFIGURATION

3.1 Hardware Requirements:

The minimum hardware requirements for developing and running the proposed system are:

- Processor: Intel Core i3 or i5 or AMD Ryzen 5
- RAM: 8 GB (Minimum 4 GB)
- Storage: 256 GB SSD or above
- Operating System: Windows 10/11, Linux, or macOS
- Internet Connection: Required for accessing Gemini API, News APIs, and Fact-Check APIs

3.2 Software Requirements:

The software requirements for the proposed system are:

- Programming Language: Python 3.10 or above
- Frontend: HTML5, CSS3, JavaScript
- Backend Framework: Flask or FastAPI
- IDE/Code Editor: Visual Studio Code
- Database: SQLite (or MySQL, if used)
- AI Model: Gemini API
- NLP Library: Sentence Transformers (SBERT)
- News Retrieval: NewsAPI
- Fact Verification: Google Fact Check API
- Libraries: Transformers, Requests, Pandas, NumPy, Scikit-learn
- Version Control: Git and GitHub
- Web Browser: Google Chrome, Microsoft Edge, or Any Browser

CHAPTER 4

MODULE DESCRIPTION

The proposed system is divided into the following modules

1. User Input Module

This module allows users to enter a news article or claim through a web interface. The input text is collected and sent to the backend for verification.

2. Claim Extraction Module (Gemini API)

This module uses the Gemini API to extract important information such as keywords, entities, and factual claims from the user's input. It helps structure the claim for further analysis.

3. News Retrieval Module

This module searches trusted news sources using News APIs and retrieves related articles that may support or contradict the user's claim.

4. Fact-Check Verification Module

This module checks professional fact-checking databases to determine whether the claim has already been verified or debunked.

5. Semantic Similarity Module (SBERT)

This module compares the user's claim with retrieved news articles using Sentence Transformers (SBERT) and calculates a semantic similarity score.

6. Source Credibility Module

This module evaluates the reliability of news sources. Trusted organizations receive higher credibility scores, while unknown or unreliable websites receive lower scores.

7. Decision Engine Module

This module combines similarity scores, source credibility, and fact-check results using rule-based logic to classify the claim as Likely Verified, Likely False, or Unverified.

8. Explanation Generator Module

This module uses Gemini to generate a simple explanation describing why the claim received its final classification and what evidence was used.

CHAPTER 5

SYSTEM DESIGN

5.1 Data Flow Diagrams

DFD Level 0: Context Diagram



Figure: 5.1.1 Data Flow Diagram

The FakeXpose system begins when a user submits a news article, headline, or claim for verification. The system processes the input and generates relevant search queries, which are sent to backend services such as the Google Search API, News API, and Gemini API to collect related news articles, search results, metadata, and AI-generated insights. FakeXpose then analyzes the retrieved information. Based on this analysis, the system determines whether the news is Verified, Likely False, Misleading, or Unverified, and returns the final verification result along with supporting evidence and an explanation to help the user understand the authenticity of the submitted news.

DFD Level 1: Context Diagram

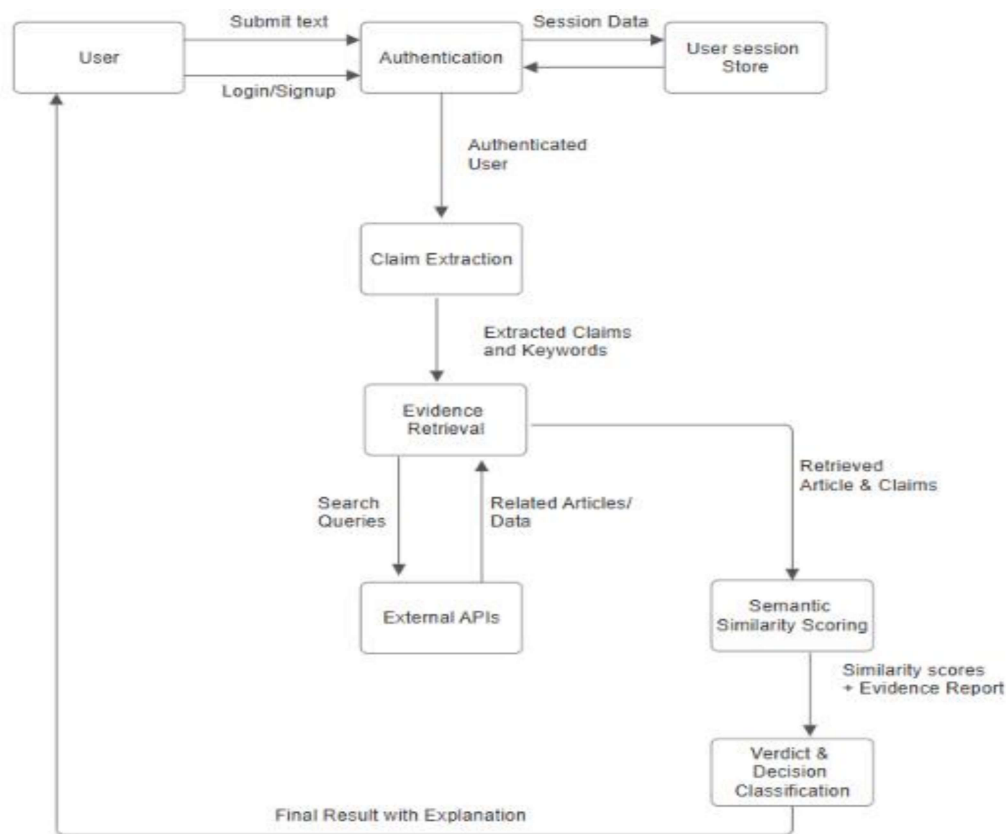


Figure: 5.1.2 Context Diagram

The workflow begins when a user logs into or signs up for the system through the authentication module, where the user's session is securely stored in the session database. After successful authentication, the user submits a news article or claim, and the Claim Extraction module identifies the main claims and important keywords. These extracted keywords are then sent to the Evidence Retrieval module, which queries external APIs and trusted news sources to collect related articles and fact-checking evidence. The retrieved information is passed to the Semantic Similarity Scoring module, which compares the user's claim with the collected evidence using natural language processing techniques and calculates similarity scores. Finally, the Verdict and Decision Classification module analyzes these scores and classifies the claim as True, False, Misleading, or Unverified, along with a detailed explanation. The final result is then displayed to the user,

providing both the classification and the supporting evidence for transparency.

5.2 Use Case Diagram

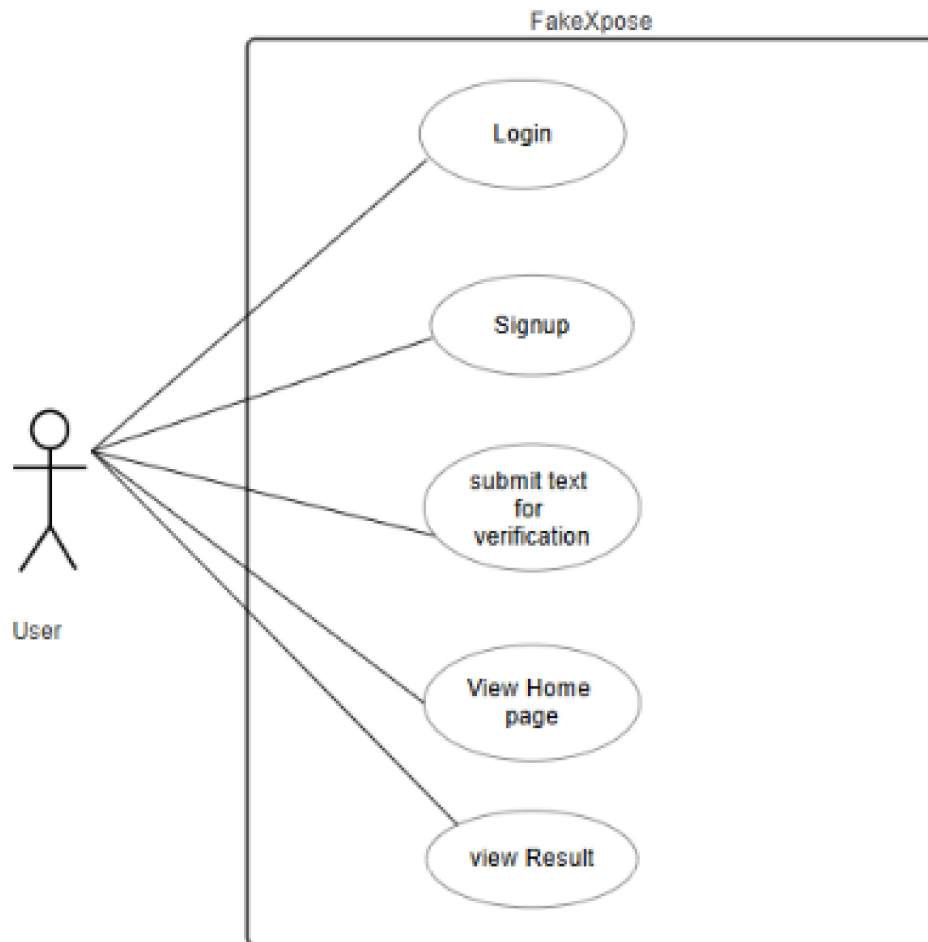


Figure: 5.2.1 Use Case Diagram

The use case diagram shows the interactions between the user and the FakeXpose system. A user can sign up, log in, access the home page, submit text for verification, and view the verification results. These use cases represent the main functionalities available to users, allowing them to verify news or claims and receive the final fact-checking outcome through the system.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 Implementation

The Fake News Detection and Verification System is implemented as a web application using Python, Flask/Fast API, HTML, CSS, and JavaScript. The frontend provides a user-friendly interface where users can register, log in, and submit a news text or claim for verification. The backend processes the request and communicates with various APIs and AI models to verify the news.

When a user submits a news article, the system first sends the text to the Gemini API, which extracts important claims, keywords, and named entities. These extracted claims are then used to search related articles from trusted news sources through the News API. The system also queries the Google Fact Check API to determine whether the claim has already been verified by professional fact-checking organizations.

The retrieved news articles are compared with the user's claim using Sentence Transformers (SBERT), which calculates the semantic similarity between the claim and the retrieved evidence. The system also evaluates the credibility of the news sources by assigning trust scores to reliable and unreliable domains.

A rule-based decision engine combines the semantic similarity score, fact-check results, and source credibility score to classify the news as Likely Verified, Likely False, or Unverified. Finally, the Gemini API generates a clear explanation describing the reason for the classification, and the result is displayed to the user through the web interface.

6.2 Screenshots



Figure: 6.2.1 Home Page

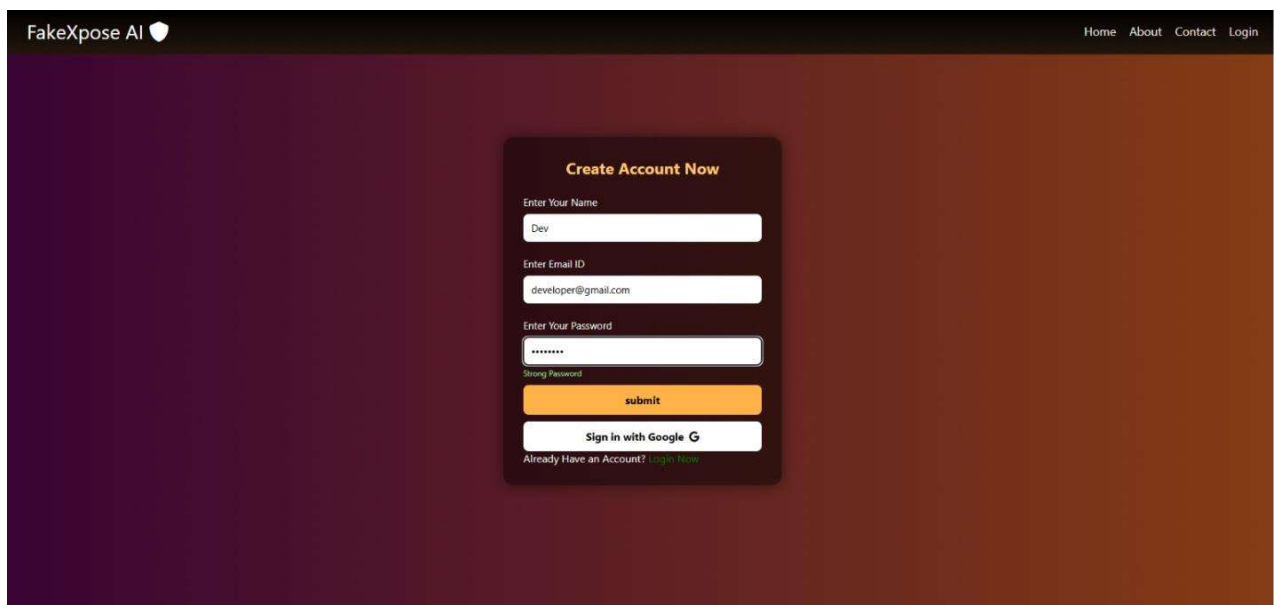


Figure: 6.2.2 Signup Page

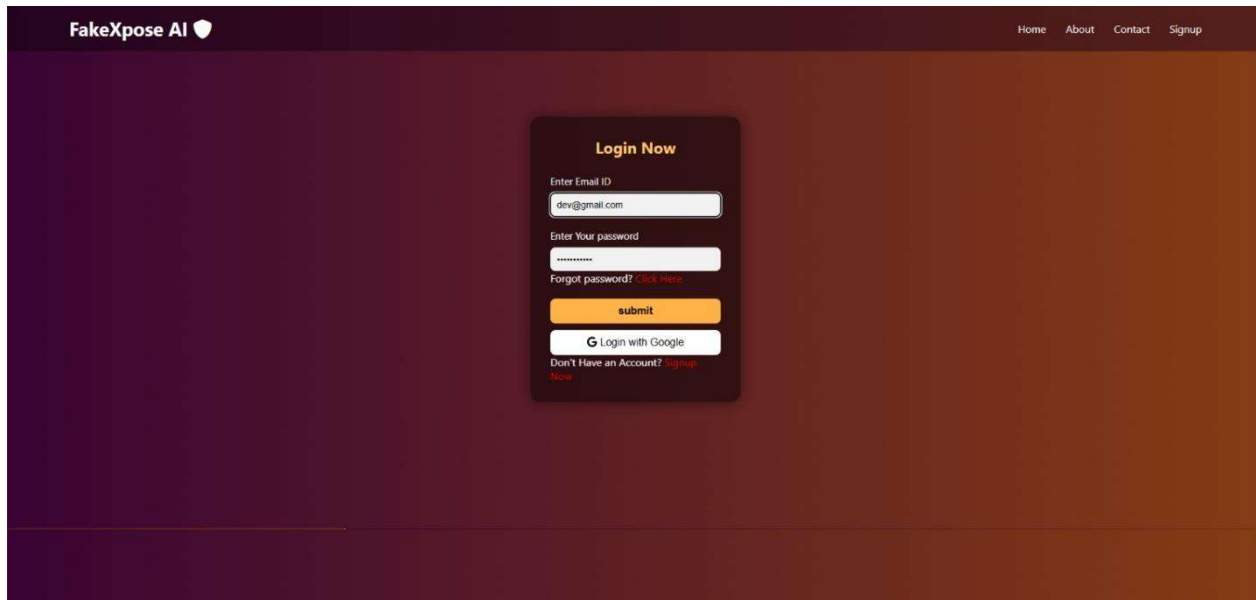


Figure: 6.2.3 Login Page



Figure: 6.2.4 Fake News Detection [Verified News]

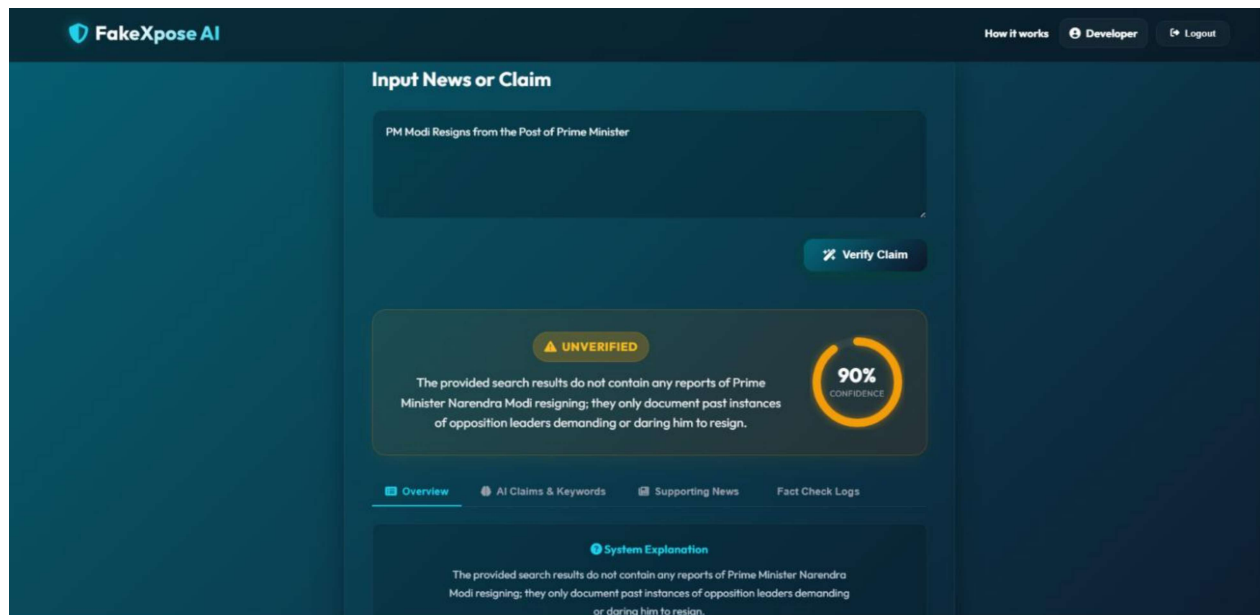


Figure: 6.2.5 Fake News Detection [Verified News]

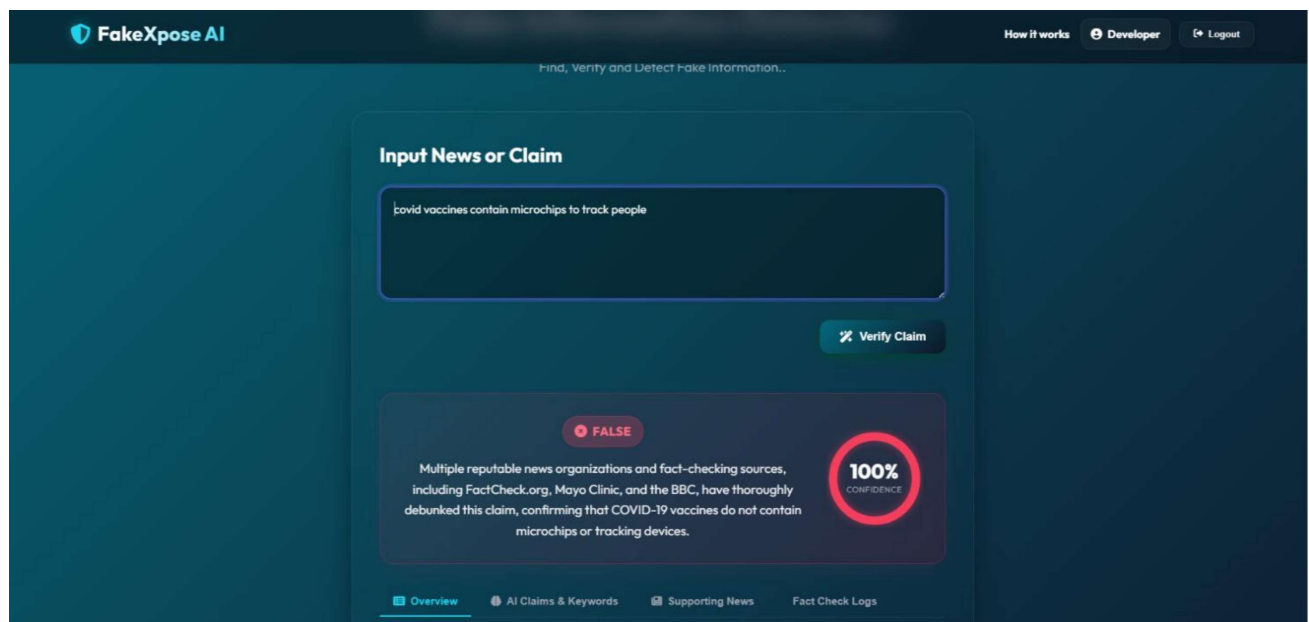


Figure: 6.2.6 Fake News Detection [False News]

CHAPTER 7

SYSTEM TESTING

The testing phase was carried out to ensure that the AI-Based Fake Information Detector functions correctly, reliably, and meets the intended system requirements.

The first test case (TC01) verifies the User Registration module by providing valid user details. The expected outcome is the successful creation of a new user account, and the system correctly registers the user, confirming that the registration functionality works properly. The second test case (TC02) evaluates the User Login process by entering a valid username and password. The system successfully authenticates the user and grants access, ensuring that only registered users can use the application.

The third test case (TC03) checks the Submit Valid News functionality by providing a genuine news article as input. The expected result is that the system processes the article for verification using AI models and external fact-checking sources. The actual result confirms that the news is processed successfully. The fourth test case (TC04) verifies the system's ability to identify Fake News. When a fake news article is submitted, the application correctly classifies it as Likely False, demonstrating the effectiveness of the credibility analysis and verification process.

The fifth test case (TC05) validates the detection of Verified News. A genuine and factually correct news article is submitted, and the system accurately classifies it as Likely Verified, indicating that the verification mechanism correctly identifies trustworthy information. The sixth test case (TC06) examines how the system handles claims that lack sufficient supporting evidence. In such cases, the application classifies the news as Unverified, ensuring that uncertain information is not incorrectly labelled as true or false.

The seventh test case (TC07) focuses on the Explanation Generation feature. After the classification process, the user requests an explanation of the result. The system successfully displays the explanation generated by the Gemini AI model, helping users understand why a particular classification was made. This feature improves transparency and increases user confidence in the system's decisions.

The eighth test case (TC08) verifies Input Validation by attempting to submit an empty news

article. Instead of processing the request, the system displays an appropriate validation message, preventing invalid input from being processed.

The ninth test case (TC09) evaluates the communication between the application and external services such as the Gemini API, News API, and Fact Check API. The expected outcome is successful data exchange with these APIs. During testing, all API requests were completed successfully, confirming that the system can retrieve and process external information required for accurate fact verification.

Finally, the tenth test case (TC10) validates the Logout functionality. When the user clicks the logout option, the application terminates the active session and redirects the user to the login page. The successful execution of this test confirms that session management and user security are implemented correctly

Test Cases

Test Case ID	Test Case	Input	Expected Result	Actual Result	Status
TC01	User Registration	Valid user details	User account created successfully	Account created successfully	Pass
TC02	User Login	Valid username and password	User logged into the system	Login successful	Pass
TC03	Submit Valid News	Genuine news article	News is processed for verification	Process completed	Pass
TC04	Submit Fake News	Fake news article	System classifies as Likely False	Classified as Likely False	Pass
TC05	Submit Verified News	Verified news article	System classifies as Likely Verified	Classified as Likely Verified	Pass
TC06	Submit Unknown Claim	News with no supporting evidence	System classifies as Unverified	Classified as Unverified	Pass
TC07	Explanation	View Explanation	Explanation is displayed	Explanation displayed successfully	Pass
TC08	Empty Input	No text entered	Display validation message	Validation message displayed	Pass
TC09	API Connection	Request to Gemini, News API, and Fact Check API	APIs respond successfully	APIs responded successfully	Pass
TC10	Logout	Click Logout	User session ends and redirects to login page	Logout successful	Pass

Test Case Description

- TC01: Verifies that a new user can successfully register.
- TC02: Ensures that only valid users can log in.
- TC03: Checks whether the system accepts and processes a valid news article.
- TC04: Confirms that fake news is correctly classified as Likely False.
- TC05: Confirms that verified news is classified as Likely Verified.
- TC06: Ensures that claims without sufficient evidence are marked as Unverified.
- TC07: Verifies that the explanation generated by Gemini is displayed correctly.
- TC08: Checks that the system prevents empty submissions.
- TC09: Ensures successful communication with external APIs.
- TC10: Verifies that the logout functionality works correctly.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Conclusions

This project presents an AI-powered Fake News Detection and Verification System that combines Artificial Intelligence, Natural Language Processing, and real-world evidence to verify the authenticity of news articles and claims. Unlike traditional fake news detection systems, the proposed system does not rely only on machine learning predictions. Instead, it uses the Gemini API for claim extraction and explanation generation, News APIs for retrieving current news, Google Fact Check API for verification, SBERT for semantic similarity analysis, and source credibility analysis to evaluate the reliability of news sources.

The rule-based decision engine combines all these components to classify news as Likely Verified, Likely False, or Unverified. The developed system provides accurate, transparent, and explainable results, helping users make informed decisions before believing or sharing online information. Overall, the project demonstrates that combining AI with real-time evidence improves the reliability of fake news detection.

8.2 Future Enhancements

The proposed system can be further improved by adding new features and technologies in the future. Some possible enhancements include:

- Integrating more trusted news and fact-checking APIs for better verification.
- Supporting multiple languages to detect fake news in regional and international languages.
- Detecting fake news from images, videos, and social media posts.
- Developing a mobile application for Android and iOS users.
- Improving the rule-based decision engine using advanced AI techniques.
- Adding browser extensions to verify news directly while browsing websites.
- Using blockchain technology to improve the authenticity and traceability of news.
- Continuously updating the source credibility database to improve verification accuracy.

CHAPTER 9

REFERENCES

9.1 Text References

- [1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", Proceedings of NAACL-HLT, 2019.
- [2] Musfikur Rahaman, "NewsLogic :AI-Powered Fake News Detection System", Research Gate, 2025
- [3] Rafael Nicholas Tanaja, Johnny, Musyaffa Ayman Rafif, and Alexander Agung Santoso Gunawan, "Fake News Detection using Machine Learning: Integrating FakeBERT Classification, Style Analysis, and Credibility Verification," 2025.
- [4] "Fake News Detection using Machine Learning and NLP with Missing Data Handling," 2023.

9.2 Web References:

- Gemini API Documentation <https://ai.google.dev/>
- NewsAPI Documentation <https://newsapi.org/>
- Google Fact Check Tools API Documentation <https://developers.google.com/fact-check/tools/api>
- Sentence Transformers Documentation <https://www.sbert.net/>
- Fake News Detection using Machine Learning <https://www.geeksforgeeks.org>
- Fake News Detection System www.ijert.org